

Users & sudo Cheat Sheet

Check who you are, add and change users and groups, and control who can run things as root.

Who am I?

```
$ id
uid=1000(mark) gid=1000(mark) groups=1000(mark),4(adm),27(sudo),998(docker)
```

uid = user number · **gid** = main group · **groups** = all your groups. In **sudo** or **wheel** = can use sudo.

Command	Shows	Command	Shows
whoami	Your user name	getent passwd alice	alice's account line
id alice	Another user's IDs and groups	getent group sudo	Who is in group sudo
groups	Just your group names	w / last	Who's logged in / recent logins

Managing users and groups (needs sudo)

Command	Does	Command	Does
useradd -m -s /bin/bash alice	New user, home folder, bash	usermod -L alice	Locks login (-U undoes)
passwd alice	Sets alice's password	userdel -r alice	Deletes alice and her home
usermod -aG docker alice	Adds alice to group docker	groupadd devs	Creates a group
gpasswd -d alice docker	Removes her from one group	passwd	Changes your own password

Watch out: **usermod -G** without **-a** **replaces** all her extra groups. Debian/Ubuntu: **adduser alice** is friendlier.

Reading /etc/passwd (one line per user; anyone can read it)

```
alice:x:1001:1001:Alice Smith,,,:/home/alice:/bin/bash
```

Field	Means	Field	Means
1 alice	User name	5 Alice Smith	Full name / comment
2 x	Password is in /etc/shadow	6 /home/alice	Home folder
3 1001	User ID (below 1000 = system account)	7 /bin/bash	Login shell (no login = can't log in)
4 1001	Main group ID		

/etc/shadow = password **hashes** and expiry, root-only (! before the hash = locked; **chage -l alice** shows expiry).

su vs sudo

Command	Does	Command	Does
sudo cmd	One command as root (your password)	su -	Become root (root's password)
sudo -i	Root login shell (your password)	su - alice	Become alice (her password)
sudo -u postgres psql	Runs a command as another user	sudo -l	What am I allowed to run?
sudoedit /etc/hosts	Edits a root-owned file safely	sudo -k	Forget my password now

Reading sudoers rules

```
%sudo    ALL=(ALL:ALL) ALL    ← group sudo: anything, as anyone, on any host
deploy   ALL=(root) NOPASSWD: /usr/bin/systemctl restart myapp ← just one command, no password
```

Pattern: **who** (user, or %group) **host=(run as user:group)** **commands** · **NOPASSWD:** = no password prompt

Edit only with **sudo visudo**: it checks syntax first, and a broken file locks everyone out of sudo. Keep your rules in their own file: **sudo visudo -f /etc/sudoers.d/deploy** (no dots in the name).

Handy recipes

Command	Does
awk -F: '\$3>=1000 && \$3<65534' /etc/passwd	Lists real (human) accounts
sudo -u www-data cat /var/www/app/.env	Tests whether a service user can read a file

Remember: always **usermod -aG**, never just **-G**. Group changes only apply to **new logins**: the user must log out and back in (or run **newgrp docker**) before **id** shows the new group.